



A novel tool for Bayesian reliability analysis using AHP as a framework for prior elicitation

Zaki Syed^{*}, Oleg Shabarchin, Yuri Lawryshyn

Centre for Management of Technology and Entrepreneurship, Faculty of Applied Science and Engineering, University of Toronto, 200 College Street Rm 256, Toronto, Ontario M5S 3E5, Canada

ARTICLE INFO

Keywords:

Bayesian reliability analysis
Quantitative risk analysis
Expert elicitation
Analytic hierarchy process
Weibull

ABSTRACT

Quantitative risk assessment (QRA) is a powerful and popular technique to support risk-based decisions. Unfortunately, QRAs are often hampered by significant uncertainty in the frequency of failure estimation for physical assets. This uncertainty is largely due to lack of quality failure data in published sources. The failure data may be limited, incompatible and/or outdated. Consequently, there is a need for robust methods and tools that can incorporate all available information to facilitate reliability analysis of critical assets such as pipelines, pressure vessels, rotating equipment, etc. This paper presents a novel practical approach that can be used to help overcome data scarcity issues in reliability analysis. A Bayesian framework is implemented to cohesively integrate objective data with expert opinion with the aim toward deriving time to failure distributions for physical assets. The Analytic Hierarchy Process is utilized to aggregate time to failure estimates from multiple experts to minimize biases and address inconsistencies in their estimates. These estimates are summarized in the form of informative priors that are implemented in a Bayesian update procedure for the Weibull distribution. The flexibility of the proposed methodology allows for efficiently dealing with data limitations. Application of the proposed approach is illustrated using a case study.

1. Introduction

Since its introduction to the chemical process industry in the late 1970s, quantitative risk assessment (QRA) has become a powerful technique to support decision-making in the areas of process and safety system design, land use planning and facility siting, and planning asset inspection and maintenance activities to name a few (Pasman and Reniers, 2014; Villa et al., 2016). Whilst several important advancements have been made to make QRAs more complete, defensible and repeatable, frequency of failure data upon which QRA relies has proven to be the most difficult to improve. Often QRAs rely on limited, generic/incompatible, and/or outdated failure data available through published reliability databases (Villa et al., 2016). The issue of limited and highly uncertain failure data is one of the primary concerns for industry and regulators, resulting in reluctance to adopt risk-based decision-making. As such, there is a recognition by advocates of the QRA method that better data and analytics is crucial for improving QRAs (Pasman and Reniers, 2014).

The advent of easily accessible information coupled with ever increasing computing power solidifies the application of predictive

analytics to facilitate efficient decision-making. Some industries, through numerous examples, have shown the advantage of using predictive analytics such as machine learning and artificial intelligence as a decision support tool (Knox, 2018). Such analytics are data centric at their core and may not require any knowledge of the underlying theory/process of a system under consideration. Thus, when good quality data is in abundance, these approaches can be employed to inform decision-making. However, despite the tremendous appetite for using such analytics, misleading results can be obtained if the analytics are based on limited, missing or poor-quality data (Redman, 2018).

Risk practitioners are acutely aware of the problems of incomplete and poor-quality data, particularly for supporting risk-based decisions related to physical asset management and process safety management. In practice, the data problems are common for many industries, including energy, oil and gas, utilities, etc. (Hong et al., 2009a,b). Nevertheless, managers looking to make risk-based decisions must do so with limited data, and thus high uncertainty. Often the decision carries with it significant cost implications, making effective decision-making evermore critical. For example, many physical assets such as pipelines and rotating equipment are expensive to maintain or replace; and

^{*} Corresponding author.

E-mail addresses: zaki.syed@live.com (Z. Syed), oshabar@gmail.com (O. Shabarchin), yuri.lawryshyn@utoronto.ca (Y. Lawryshyn).

engineering of process design changes or installation and maintenance of safety systems can be costly. As such, a method to incorporate all available information about an asset is necessary to facilitate decision-making when faced with data scarcity.

The Bayesian paradigm offers a mathematically consistent way of augmenting the available data with other sources of information, and in turn reducing the amount of uncertainty faced by a decision-maker. In this paper, we apply Bayesian methods in reliability analysis of physical assets. Much of the research in reliability of engineered systems or components has successfully integrated multiple sources of information by applying informative prior distributions within the Bayesian paradigm. Wilson and Fronczyk (2016) showed how prior information can improve precision in reliability calculations for military application purposes. The authors applied laboratory test results to specify an informative prior, which was then used to supplement the experimental data. Wu (2017) implemented an outcome of fatigue analysis to rigorously construct informative priors in a Bayesian Hierarchical model for reliability of wind turbines. Such integration of prior information enabled reliability of engineered systems or components to be estimated even under significant data scarcity.

Li and Meeker (2014) compared multiple cases of using informative prior, non-informative prior and maximum likelihood (MLE) methods for reliability studies. While non-informative and traditional MLE methods provide almost identical results, informative priors, based on accelerated degradation tests, significantly contributed to uncertainty reduction in all reliability case studies. Furthermore, the authors argue that explicitly defined informative priors establish a better foundation for decision-making, due to its inherent transparency. Other researchers advocate that properly specified priors (even weakly informative) not only help to formally integrate various types of information, but also help to minimize convergence issues for numerical algorithms, such as Markov Chain Monte Carlo (MCMC) (McNeish, 2016; Lambert, 2018).

It is not always possible to obtain objective prior distributions from physics of failure simulations or accelerated life test results, as these methods can be costly. When resource constraints preclude objective prior information, subjective prior knowledge elicited from experts may be used. In this paper we present a methodology to incorporate expert opinion with actual failure data with the goal of obtaining a Weibull life distribution for a physical asset. The Analytic Hierarchy Process (AHP) is implemented to provide a structured approach for selecting experts from whom to elicit time to failure estimates (Saaty, 1980). AHP is also used to aggregate opinions of multiple experts to construct informative priors.

A similar method was suggested by Cagno et al. (2000). The authors used AHP in the Bayesian paradigm to obtain prior distributions for the failure rate of different low-pressure cast-iron gas pipelines. However, AHP was utilized for pairwise comparison of assets under different operating conditions rather than for expert selection and aggregation of expert opinion. AHP was used by Fengqiang et al. (2006) to obtain weights for the purpose of merging multiple prior distributions. While expert opinion was used to derive the weights through AHP, the authors did not use AHP as a framework to elicit the prior distributions, as we have done in this work.

The use of expert opinion to construct prior distributions of Weibull parameters has been explored by authors in the past (Singpurwalla, 1988; Coolen et al., 1990; Gutierrez-Pulido et al., 2005). However, such methods have relied on experts to provide estimates about the values of unobservable quantities such as the Weibull shape parameter, which can be a difficult task for experts. The method presented in this paper requires experts to provide estimates on time to failure, which is an observable quantity and thus easier for the experts to estimate. Time to failure estimates are then used to obtain prior distributions for Weibull parameters.

We begin by providing a summary of the theoretical background in Section 2. A detailed description of the methodology is provided in Section 3. Finally, we demonstrate the utility of the proposed

methodology using a case study based on actual experience with natural gas compression and transmission infrastructure (Section 4).

2. Theoretical background

2.1. Reliability engineering

The concept of reliability for engineered (physical) assets is best understood by first considering the meaning of quality, which is the set of characteristics of an asset that determines if it meets specifications and in turn determines customer satisfaction. The reliability of an asset is its ability to retain its quality as it ages. Mathematically, the reliability (R) of an asset is defined as the probability that it continues to retain its quality over a given period of time, whilst operating in a given environment (Bentley, 1999). Conversely, the unreliability (F) of an asset is the probability that it fails to meet specifications.

The unreliability of an asset as function of age (t) is given by the cumulative distribution function $F(t)$, representing the probability that the asset fails by age t . The reliability of an asset is given by $R(t) = 1 - F(t)$ and the instantaneous failure rate (also known as the hazard rate) is given by $h(t) = f(t)/R(t)$, where $f(t)$ is the probability density function of time to failure (Tobias and Trindade, 2012). The hazard rate is indicative of the failure mechanism: a decreasing hazard indicates early failure mechanisms such as installation or quality issues; an increasing hazard is typical of end of life or wear-out failures; and a constant hazard is indicative of no significant wear-out or early failure mechanism (Jardine and Tsang, 2013). The age at which an asset fails is known as the lifetime of the asset, and the uncertainty about the lifetime of an asset is represented using mathematical models known as life distributions. One of the most common models is known as the Weibull distribution, defined by the density and distribution functions

$$f(t; \beta, \eta) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right] \quad \text{and} \quad F(t; \beta, \eta) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right], \quad (1)$$

where $\eta > 0$ is the scale parameter and $\beta > 0$ is the shape parameter, and $t \geq 0$ (Tobias and Trindade, 2012). The Weibull distribution has gained widespread use because of its flexibility, allowing it to fit many failure characteristics. When $\beta = 1$, the Weibull distribution exhibits constant hazard. When $\beta < 1$, the Weibull distribution exhibits decreasing hazard. When $\beta > 1$, the Weibull distribution exhibits increasing hazard.

Typically, parameters are estimated using regression methods (Weibull analysis) or the method of maximum likelihood (MLE) (Jardine and Tsang, 2013). Such methods are classical approaches to statistical analysis, and they perform well when sample sizes are high (Tobias and Trindade, 2012). In practice, a sufficient sample is often not available for multiple reasons. For example, assets may operate in vastly different operating environments or are of different design. This would require stratification, reducing the number of samples in each group of assets. Another possibility is that assets may be highly reliable resulting in limited failures. Or data collection may have been flawed or occurred over a limited duration, resulting in few samples that can be used for statistical analysis. Bayesian statistical inference can be used to fill this gap and can yield more information about the asset life than classical methods if used appropriately (Tobias and Trindade, 2012).

2.2. Bayesian inference in reliability analysis

Bayesian updating of an initial hypothesis is performed using Bayes' theorem. The theorem describes how to update a prior belief about a phenomenon or process considering new evidence, and is given by

$$p(\theta|x) = \frac{p(x|\theta) p(\theta)}{p(x)} = \frac{p(x|\theta) p(\theta)}{\int p(x|\theta) p(\theta) d\theta}, \quad (2)$$

where $p(\theta|x)$ is the posterior probability distribution for the set of parameters (θ) describing the process and x is the observed data. The prior

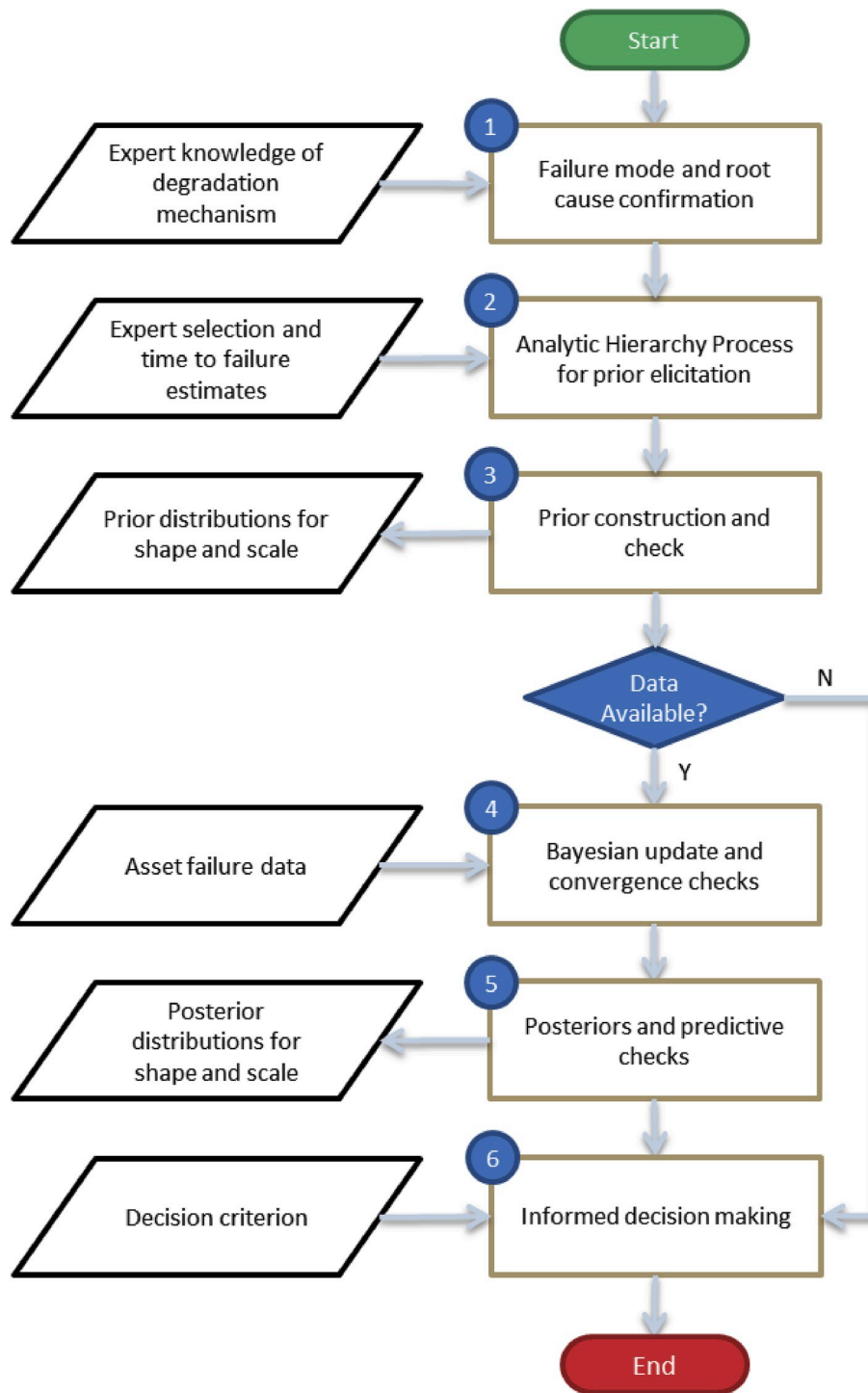


Fig. 1. Decision making flow chart.

distribution $p(\theta)$ reflects the initial state of knowledge about the process before observing any data, whereas the posterior distribution $p(\theta|x)$ shows the updated state of knowledge after new evidence becomes available. These two expressions are linked through a likelihood function $p(x|\theta)$. This function describes the plausibility of the observed data given model parameters θ . Lastly, $p(x)$ is the marginal distribution of the data and is given by $\int_{All \theta} p(x|\theta) p(\theta) d\theta$ (Hamada et al., 2008).

In the literature, the prior is also often referred as a subjective distribution. Some degree of subjectivity may be involved to construct a prior distribution. This subjectivity is often used as a basis for criticism

of Bayesian methods. Despite this criticism, the prior distribution is not a shortcoming, rather a key feature of the Bayesian paradigm. The prior distribution offers a flexible and cohesive approach to incorporate different types of information. This property is particularly useful in reliability studies where a significant amount of prior information and tacit knowledge is often available but cannot be incorporated using classical methods of reliability analysis. Prior knowledge may include, but is not limited to, asset condition information, engineering judgement, accelerated life testing, industry failure data on similar assets, and so on. Thus, incorporation of such information through prior distributions helps to make more realistic assumptions as opposed to fully

relying on data alone and assuming no other information is available (Li and Meeker, 2014). Furthermore, the issue of data availability and small sample size in general is highly relevant for reliability studies, especially in the modeling of rare events. In fact, for rare events, one may need to rely more on engineering knowledge and tacit information than on actual scarce incident data (Sridhar, 2018).

To ensure superiority of Bayesian inference, the prior distribution must be rigorously constructed. This may not be crucial for a large sample size (i.e. data overwhelms importance of prior), but for a small sample size a properly specified prior is essential because the choice of prior may have a predominant impact on the posterior (McNeish, 2016). As noted by van de Schoot et al. (2013): “the smaller the sample size, the bigger the influence of the prior specification and the more can be gained from specifying subjective priors”. Thus, for small sample size problems as much advantage can be gained by using Bayesian methods, as much bias can be introduced if the prior is incorrectly specified. To avoid the problem, this paper implements a rigorous elicitation approach to properly construct subjective prior distributions. Details on this approach are discussed in the following section.

3. Methodology

This section provides details of the proposed methodology for conducting reliability analysis using Bayesian methods when faced with limited failure data. As discussed previously, Bayesian methods for reliability analysis have a distinct advantage over classical approaches through the use of informative prior distributions for model parameters. The methodology presented in this paper provides a framework for obtaining prior distributions for the Weibull distribution parameters, which are then updated using actual failure data. A flow chart of the methodology is shown in Fig. 1.

Emphasis is placed on a Weibull model for asset failure due to a wear out failure mechanism. However, this methodology is flexible and may be extended to other life distributions. The methodology first requires that the failure mechanism be understood to determine if a Weibull wear out model is applicable (Section 3.1). Once this is confirmed, AHP is used to aggregate time to failure estimates from multiple experts (Section 3.2). The expert estimates are used to obtain prior distributions for the Weibull parameters as described in Section 3.3. If failure data is available, a Bayesian update of the priors is conducted using the Metropolis-Hastings (MH) MCMC algorithm as discussed in Section 3.4. Recommended posterior checks are discussed in Section 3.5 to evaluate model fit.

In presenting this methodology, we refer to three key roles in the prior elicitation, Bayesian inference and decision-making process (Bedford et al., 2006):

- **Decision-maker:** person responsible for making the final decision (e.g. proactive replacement of assets) and must be informed about relevant uncertainties by experts and available data.
- **Experts:** persons who are subject matter experts with respect to the problem (i.e. asset reliability) of interest.
- **Analyst:** person responsible for selecting experts, eliciting their opinion with respect to the problem, and performing reliability analysis with available data.

3.1. Failure mechanism

To fully characterize a failure mechanism, it is necessary to define the failure mode as well as the root cause. This concept is perhaps best explained with an illustrative example. Consider a manually operated ball valve in a chemical process facility. A critical failure mode of this type of asset may be internal leakage. One root cause of this failure mode may be wearing of the seals due to repeated cycling (open/close) of the valve, and a different root cause may be an overturning of the valve

resulting in a misalignment between the valve stem and the ball. In the first case, the time to failure may be modeled using a Weibull distribution with $\beta > 1$ (wear out). In the second case, the failure is due to a random event that is not age dependent and may be modeled using a Weibull distribution with $\beta = 1$.

As highlighted with the above example, in order to ensure that estimates provided by experts are consistent with the failure mechanism to be modeled, the failure mode and root cause must be established before applying the proposed methodology. Furthermore, the failure mechanism must be communicated to all experts before eliciting their opinion, with the aim towards managing cognitive biases (Quigley et al., 2008). It has been proposed that qualitative structuring of the model is equally important to quantitative assessment to ensure proper parameterization of the reliability model (Bedford et al., 2006). If the failure mechanism is confirmed to exhibit a wear out pattern, the time to failure can be modeled using a Weibull distribution with $\beta > 1$, and in turn one with a non-zero mode. This characteristic, as explained in Section 3.3, is used to obtain the prior distributions for the Weibull shape and scale.

3.2. Expert elicitation

Expert knowledge and engineering judgement are essential components in understanding and estimating reliability of assets. Such information represents subjective knowledge which is encoded in the form of meaningful mathematical statements (i.e. eliciting) (Evans, 1989). This subsection presents a methodology for eliciting subjective information about asset time to failure that is then summarized in the form of informative prior distributions for the Weibull model parameters. Construction of the informative prior is a challenge, but if performed properly, it can significantly improve inference, especially in the case of a small sample of actual asset failures. AHP is used to aggregate expert estimates in an objective and structured manner.

AHP is a multi-criteria decision-making methodology that has been extensively studied and widely used in many industries (Saaty, 1980). The objective of AHP is to provide decision-makers with a structured method of evaluating alternative solutions against decision criteria. Alternatives can be potential capital projects, candidates for hire, equipment purchase options, etc. Decision criteria are factors that are of importance in the selection of an alternative. For example, cost, functional performance criteria, lead time, etc. The aim then is to obtain relative ranking of the solution alternatives based on how well they meet the decision criteria to help select the best alternative. AHP has been used to objectively select students for admission to graduate business school programs (Saaty and Vargas, 2012), as well as for recruitment of sales personnel (Duben and Ricks, 2015). Similarly, in this application, AHP is used to objectively select experts and then aggregate their input regarding prior information about the Weibull parameters.

We implement the AHP method with a single level in the decision hierarchy and three criteria: years of experience, number of observed failures, and level of training with respect to maintenance/operation of the asset in question. However, the methodology may be extended to any number of levels and criteria. The analyst may choose to define the AHP criteria differently depending on the problem at hand. For example, number of observed failures can refer to only observations of the exact type of asset being analyzed or may be defined to include an expert's observations of failure of similar assets. The important part is ensuring consistency in definition when performing pairwise comparison of the decision criteria and when evaluating each expert against the criteria.

The analyst, perhaps through collaboration with the decision-maker, must first select (by pairwise comparison) the relative importance of each of the decision criteria. Saaty's numeric scale is recommended (Saaty, 1980) to quantify the relative importance (Table 1). The relative importance must be evaluated based on the analyst's judgment about how much each criterion will contribute to the validity of asset time to failure estimates provided by an expert. For example, if it is known that the failure mechanism in question has historically been managed

Table 1
AHP numeric scale with linguistic interpretation.

Intensity of Importance	Definition	Explanation
1	Equal importance	Two factors contribute equally to the objective.
3	Somewhat more important	Experience and judgement slightly favour one over the other.
5	Much more important	Experience and judgement strongly favour one over the other.
7	Very much more important	Experience and judgement very strongly favour one over the other. Its importance is demonstrated in practice.
9	Absolutely more important	The evidence favouring one over the other is of the highest possible validity.
2,4,6,8	Intermediate values	When compromise is needed.

through regular inspection/maintenance then the expert's relevant training may receive high importance. Conversely, if the asset is highly reliable then the analyst may wish to put more importance on observed failures. After completing the pairwise comparison, the AHP consistency check must be performed to ensure consistency in the importance ratings.

It must be noted that for Bayesian inference to be valid, observed failures in the expert selection context must be independent of the recorded failure data to be used in Bayesian updating (Section 3.4). In other words, the expert must be instructed to form his/her opinion based on his/her personal knowledge of past, and likely unrecorded, failures. If the analyst judges that the expert's opinion is being influenced by the failure data to be used for Bayesian updating, then it is recommended that the expert be removed from the selection process or given a low score when evaluated against this criterion.

The next step requires that the analyst score the relative degree to which each expert meets the requirements of each decision criterion. It is recommended that Saaty's 1 to 9 numeric scale be used, and the analyst give a score of 9 to the expert that best meets a given criterion. All other experts are then ranked relative to the best expert. Note that rankings are done within each decision criterion, independent of an expert's ranking in another criterion. The AHP calculation is then performed by constructing the comparison matrices, calculating criteria weights, priority vectors of experts within each decision criterion, and finally the overall weighting of each expert (Saaty and Vargas, 2012). The calculated weights are used in two different ways: to create a shortlist of experts from whom to elicit time to failure estimates, and to aggregate expert estimates of asset time to failure for prior construction.

Before eliciting expert estimates and aggregating their input, the

analyst must take precautions to minimize the impact of cognitive biases. Behavioural psychology studies have shown that experts who were asked to directly evaluate a probability number of a certain event fail to provide an adequate estimate (Tversky and Kahneman, 1982). This is confirmed by Bedford and Cooke (2001) who emphasize that an expert is unlikely to provide a reliable subjective judgment of an unobservable quantity. Furthermore, expert estimates are significantly influenced by a variety of biases that human judgment is inherently subjected to (Tversky and Kahneman, 1982). To name a few: group bias, authority bias, anchoring bias and over confidence (e.g. Dunning-Kruger effect) are biases that can drastically affect the elicitation process (Johnson et al., 2010). Therefore, to ensure quality of subjective probabilities (i.e. informative priors), not only is a structured procedure necessary for proper expert elicitation, but an efficient approach to minimize impacts of biases is also required. We propose the following method for eliciting expert estimates:

- Avoid eliciting estimates in a group and opt for individual interviews instead. The aim is to eliminate authority bias and group bias. If time/resource restrictions put a limit on the number of interviews that can be conducted, select the top experts as ranked by the AHP methodology proposed.
- Ensure that the expert is fully aware of the failure mechanism being examined and that there is minimal ambiguity or room for interpretation (see Section 3.1).
- Minimize the effect of anchoring bias and overconfidence by involving multiple experts, whose opinions are aggregated using the weights derived from the AHP methodology proposed.
- Ask the expert for directly observable quantities such as the median or mode of the asset times to failure rather than percentiles, means or variances. Suggested questions for modeling wear out failures using the Weibull distribution are discussed in Section 3.3.

3.3. Prior construction

A method for construction of prior distributions for the Weibull parameters is discussed in this section. The objectives are two-fold: capture the expert's uncertainty about the observable quantities and construct priors using expert estimates of observable quantities. A Weibull distribution used to model time to failure due to a wear out mechanism exhibits a non-zero mode. This characteristic was used by Lad and Kulkarni (2010) to obtain Weibull parameters from expert estimates of time to failure. The mode (T_{mode}) of the time to failure distribution is an observable quantity and can be interpreted as the age at which an asset typically fails or is most likely to fail. Lad and Kulkarni (2010) suggest obtaining a second observable quantity; the maximum survival age of

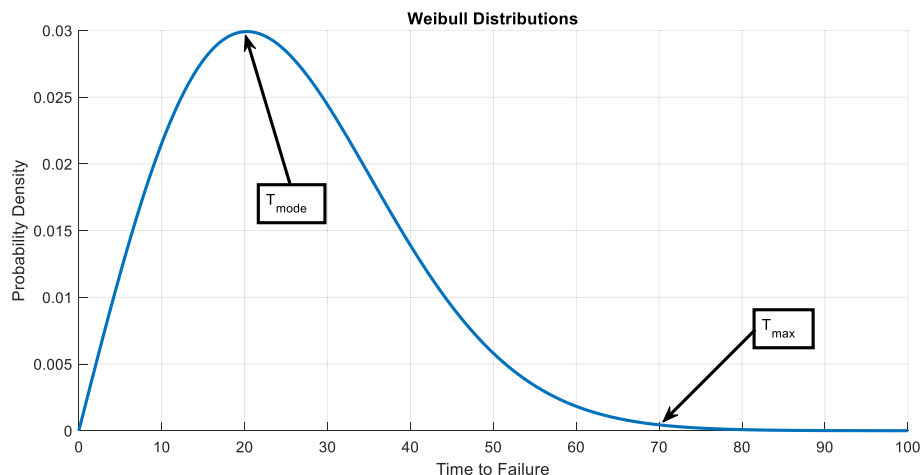


Fig. 2. Example weibull distribution.

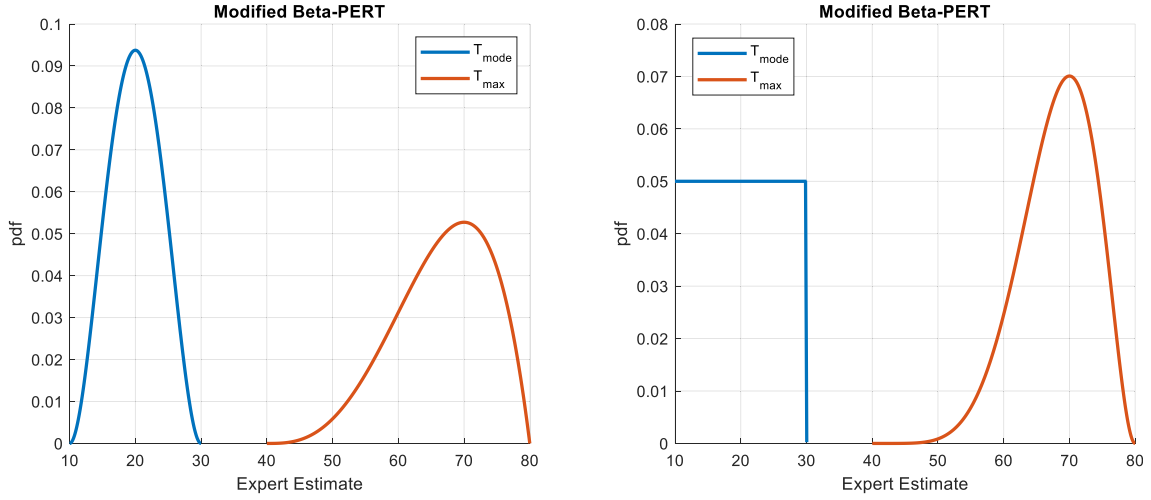


Fig. 3. Example Modified Beta-PERT distributions for $\bar{T}_{mode}$ and $\bar{T}_{max}$. In both charts: $\bar{T}_{mode}^{LB} = 10$, $\bar{T}_{mode}^{UB} = 30$, $\bar{T}_{mode}^{BE} = 20$, $\bar{T}_{max}^{LB} = 40$, $\bar{T}_{max}^{UB} = 80$ and $\bar{T}_{max}^{BE} = 70$. In left chart $\gamma = 4$ for both distributions. In right chart, $\gamma = 0$ is used for $\bar{T}_{mode}$ and $\gamma = 8$ is used for $\bar{T}_{max}$.

the asset (T_{max}) defined as the 99th percentile of the Weibull time to failure distribution. The expert may be prompted by asking for the age of the longest surviving asset. An example Weibull distribution with $T_{mode} = 20$ and $T_{max} = 70$ is shown in Fig. 2.

To capture the expert's uncertainty about T_{mode} and T_{max} we suggest prompting him/her for a lower bound ($\bar{T}_{mode}^{LB}$ and $\bar{T}_{max}^{LB}$), best estimate ($\bar{T}_{mode}^{BE}$ and $\bar{T}_{max}^{BE}$) and upper bound ($\bar{T}_{mode}^{UB}$ and $\bar{T}_{max}^{UB}$). Estimates should be elicited from multiple experts and a weighted average of each quantity ($\bar{T}_{mode}$ and $\bar{T}_{max}$) calculated using the weights obtained via AHP as described in Section 3.2 and illustrated with an example in Section 4. The aggregated expert estimates are used to construct Modified Beta-PERT distributions for $\bar{T}_{mode}$ and $\bar{T}_{max}$. Such distributions are ideally suited to model uncertainty based on expert opinion because of the convenient parameterization relying on four parameters: a minimum (a), mode (m), maximum (b) and a weighting factor (γ) for the mode (Vose, 2008). The pdf of the modified Beta-PERT is given by

$$f(x) = \frac{1}{B(\alpha_1, \alpha_2)} \frac{(x-a)^{\alpha_1-1} (b-x)^{\alpha_2-1}}{(b-a)^{\alpha_1+\alpha_2-1}}, \quad (3)$$

where $B(\alpha_1, \alpha_2)$ is the beta function, $\alpha_1 = 1 + \gamma \left(\frac{m-a}{b-a} \right)$, $\alpha_2 = 1 + \gamma \left(\frac{b-m}{b-a} \right)$, and $a \leq m \leq b$. The support of the density function (where it is non-zero) is $[a, b]$.

We obtain a Beta-PERT distribution for $\bar{T}_{mode}$ by setting $a = \bar{T}_{mode}^{LB}$, $b = \bar{T}_{mode}^{UB}$ and $m = \bar{T}_{mode}^{BE}$. The standard Beta-PERT distribution uses $\gamma = 4$, though higher or lower values may be used to modify the curve, where lower values have the effect of flattening the curve and higher values make it peakier (Vose, 2008). A distribution for $\bar{T}_{max}$ is similarly obtained. Example distributions are shown in Fig. 3, where $\bar{T}_{mode}^{LB} = 10$, $\bar{T}_{mode}^{UB} = 30$, $\bar{T}_{mode}^{BE} = 20$, $\bar{T}_{max}^{LB} = 40$, $\bar{T}_{max}^{UB} = 80$ and $\bar{T}_{max}^{BE} = 70$. The values used to plot Fig. 3 have been selected arbitrarily for illustration purposes. In practice the analyst would derive the values as described earlier in this subsection.

Fig. 3 shows the flexibility of the Modified Beta-PERT distribution, where selection of the weighting factor (γ) can change the shape from extremely peaky to uniformly flat. The chart on the left uses the standard value of $\gamma = 4$. In the chart on the right, $\gamma = 0$ is used for $\bar{T}_{mode}$ and $\gamma = 8$ is used for $\bar{T}_{max}$. While the introduction of this factor is a source of subjectivity, it is nevertheless a useful feature because it allows the analyst to adjust the distribution precision based on his/her judgment of the quality (i.e. accuracy) of the expert input on an absolute basis. In contrast, the AHP weights provide a measure of quality based on relative weight of experts to each other.

Once distributions for $\bar{T}_{mode}$ and $\bar{T}_{max}$ have been constructed, an algorithm is used to generate distributions for the Weibull parameters (β and η). The algorithm randomly samples a value of $\bar{T}_{mode}$ and $\bar{T}_{max}$ from the Beta-PERT distributions of each and then calculates the Weibull

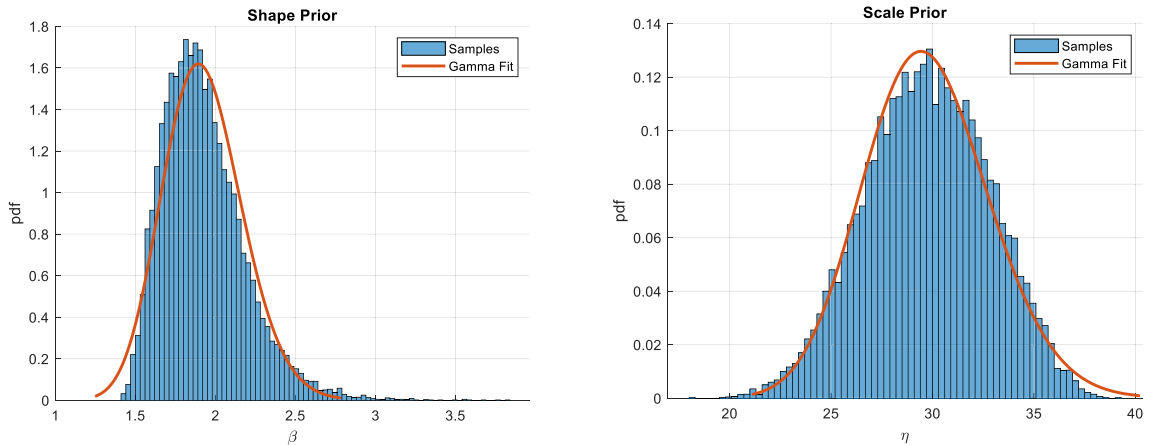


Fig. 4. Gamma fit for Weibull parameter samples.

parameters for the pair of samples. The Weibull parameters are calculated by solving equations for $\bar{T}_{mode}$ and $\bar{T}_{max}$ simultaneously as proposed by [Lad and Kulkarni \(2010\)](#). The maxima of the Weibull pdf can be found by simple differentiation of $f(t)$ with respect to t (see Equation (1)) and is given by

$$\bar{T}_{mode} = \eta \left[\frac{\beta - 1}{\beta} \right]^{\frac{1}{\beta}}. \quad (4)$$

An expression for $\bar{T}_{max}$ is found by setting the Weibull CDF ($F(t)$) equal to 0.99 and re-arranging to give

$$\bar{T}_{max} = \eta \left[\ln \left(\frac{1}{1 - 0.99} \right) \right]^{\frac{1}{\beta}}. \quad (5)$$

Equations (4) and (5) can be solved for η and β . We generate sample data for each Weibull parameter in this manner and proceed to fit gamma distributions to the sample data (Fig. 4). The parameterization of the gamma distribution commonly used in Bayesian statistics is used:

$$f(x) = \frac{\left(\frac{\mu^2}{\sigma^2} \right)^{\frac{\mu^2}{\sigma^2}} \exp \left(-\frac{\mu^2}{\sigma^2} x \right) x^{\frac{\mu^2}{\sigma^2} - 1}}{\Gamma \left(\frac{\mu^2}{\sigma^2} \right)}, \quad (6)$$

where μ is the mean and σ^2 is the variance ([Lambert, 2018](#)). Fig. 4 shows examples of priors constructed by sampling $\bar{T}_{mode}$ and $\bar{T}_{max}$ from the distributions shown in the left-hand chart in Fig. 3. Testing with different Beta-PERT distributions for $\bar{T}_{mode}$ and $\bar{T}_{max}$ shows that the gamma distribution provides acceptable fits for sample data of Weibull shape and scale generated using the algorithm described in this subsection. Moreover, it has the appropriate support since both $\beta > 0$ and $\eta > 0$. The informative priors constructed using the method described in this subsection are used to perform Bayesian inference with actual failure data as described in Section 3.4.

3.4. Bayesian update

This subsection provides a detailed description of the Bayesian updating algorithm implemented for the methodology presented in this paper. The likelihood function ($p(x|\theta)$) is discussed (see Equation (2)). We then discuss the numerical method employed to solve Equation (2) for parameters of the Weibull distribution. This is followed by a brief discussion on convergence criteria for the implemented numerical algorithm.

Under the Bayesian paradigm, observed data (e.g. failures) is incorporated in the analysis through the likelihood function. It is not uncommon for failure data to have observational constraints, and omitting such constraints may introduce significant sampling bias ([Emura and Shiu, 2014](#); [Hong et al., 2009a,b](#)). In reliability studies, the most frequently encountered observational constraints are Left Truncation (LT), Right Censoring (RC), or a combination of both (LTRC). The LT constraint occurs when failures may have been excluded from the sample because some of the assets potentially failed before observations were started to be recorded. The RC constraint occurs when an asset is still in service when observations end or if an asset is removed for a reason other than failure. Consequently, if asset data is limited by LT as well as RC constraints simultaneously such data is called LTRC. Let x represent the set of observed times to failure (i.e. $x \in t_i$), then the likelihood function, accounting for LTRC data, proposed by [Mitra \(2012\)](#):

$$p(x|\theta) = \prod_{i \in S} \{f(t_i; \theta)\}^{\delta_i \nu_i} \times \{1 - F(t_i; \theta)\}^{(1 - \delta_i) \nu_i} \times \left\{ \frac{f(t_i; \theta)}{1 - F(\tau_i; \theta)} \right\}^{\delta_i (1 - \nu_i)} \times \left\{ \frac{1 - F(t_i; \theta)}{1 - F(\tau_i; \theta)} \right\}^{(1 - \delta_i)(1 - \nu_i)} \quad (7)$$

where $f(t_i; \theta)$ and $F(t_i; \theta)$ denote the pdf and CDF (see Equation (1)) of a single observation of time to failure t_i ; τ_i represents truncation time; ν_i is an indicator function for LT cases; δ_i is an indicator function for RC cases. The truncation time is the difference between the age at failure or censoring and the age when observations were started to be recorded. The likelihood function can be easily extended to account for other types of observational constraints such as interval-censoring, right truncation and so on ([Mitra, 2012](#)).

The outcome of Bayesian updating is the posterior distribution ($p(\theta|x)$), which is nothing else but the normalized product of the prior and likelihood, where the normalizing constant is $p(x)$ (see Equation (2)). For many cases, including for Weibull likelihood, there is no analytical solution for the integral needed to obtain $p(x)$. Numerical methods must be employed to address this problem. In this study, we use the MH MCMC sampling algorithm to solve Equation (2). The pseudocode of this algorithm is provided below, whereas detailed discussion of this approach can be found in ([Chib and Greenberg, 1995](#)).

MH MCMC Algorithm:

1. Start:
 - Set $m = 1$
2. Initialize Markov Chain:
 - Set $n = 1$
 - Generate an initial value of parameter(s) θ_n
3. Propose Next Step:
 - Generate a proposal θ_{n+1} from jumping distribution $J(\theta_{n+1}|\theta_n)$
 - Calculate acceptance ratio $r = \min \left(1, \frac{p(\theta_{n+1})}{p(\theta_n)} \frac{p(x|\theta_{n+1})}{p(x|\theta_n)} \frac{J(\theta_n|\theta_{n+1})}{J(\theta_{n+1}|\theta_n)} \right)$
4. Accept or Reject:
 - Generate a uniform random number $u \in [0, 1]$
 - If $r \geq u$ accept the new state θ_{n+1}
 - If $r < u$ reject the new state and copy the previous state $\theta_{n+1} = \theta_n$
5. Increment Markov Chain:
 - Set $n = n + 1$
 - Go back to Step 3 or if maximum iterations (N) for chain reached go to Step 6
6. End of Chain:
 - Set $m = m + 1$
 - Go back to Step 2 or end simulation if maximum number of chains (M) reached

Running the MH MCMC sampler for multiple chains and enough iterations for each chain numerically recreates the posterior distribution that is used in Bayesian inference. Once the algorithm has run through the specified number of iterations, we apply the Gelman-Rubin convergence criterion to ensure convergence of the numerical method. The Gelman-Rubin criterion compares local chain variance with global variance ([Gelman and Rubin, 1992](#)). If the difference is insignificant, then MH MCMC sampler has reached the stationary posterior distribution. The Gelman-Rubin convergence criterion is given by

$$W = \frac{1}{M} \sum_{m=1}^M s_m^2; B = \frac{N}{M-1} \sum_{m=1}^M (\bar{\theta}_m - \bar{\theta})^2; \hat{R} = \sqrt{\frac{W + \frac{1}{N}(B - W)}{W}} \quad (8)$$

where W is the within chain variance calculated from the sample variance $s_m^2 = (1/N - 1) \sum_{n=1}^N (\theta_{nm} - \bar{\theta}_m)^2$ of chain m . θ_{nm} is the n th sample of the m th chain and $\bar{\theta}_m$ is the mean of the m th chain. B is the between chain variability and $\bar{\theta}$ is the overall sample mean. $\hat{R}$ is the Gelman-Rubin convergence criterion that tends towards 1 as the simulation converges. Generally, a $\hat{R} < 1.1$ is required for convergence ([Lambert, 2018](#)).

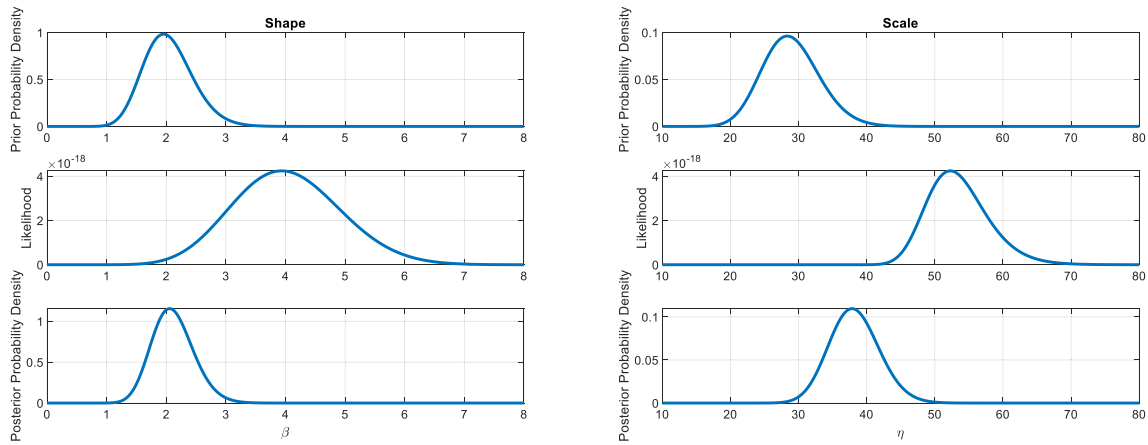


Fig. 5. PLP Plot for Weibull Parameters – 10 samples.

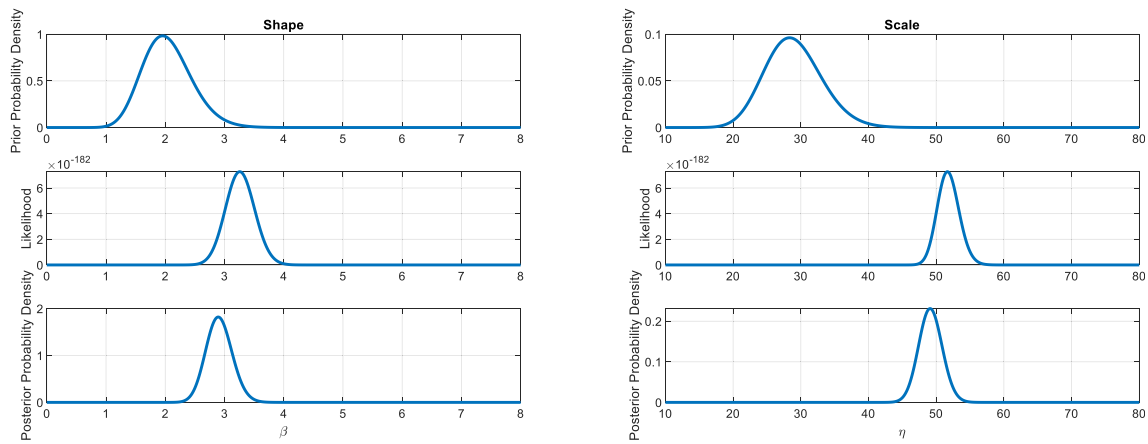


Fig. 6. PLP Plot for Weibull Parameters – 100 samples.

3.5. Posterior check

In this section we discuss methods to check the posterior distributions for η and β . We also demonstrate a visual check for the influence of the priors and likelihoods using prior-likelihood-posterior (PLP) plots. Informative priors (see example in Fig. 4) are used for Bayesian updating. Likelihoods were calculated using sample data simulated from a Weibull distribution with known parameters ($\eta = 50$ and $\beta = 3$). Parameters were arbitrarily selected for illustration purposes. The use of

selected posterior checks is demonstrated through a case study in Section 4.

Contribution of the priors and likelihoods to the posteriors can be examined visually using PLP plots. The analyst can use PLP plots to examine the degree to which the posterior is based on expert estimates versus actual data. Fig. 5 shows the PLP plot with Bayesian updating performed using 10 samples generated from a Weibull distribution with $\eta = 50$ and $\beta = 3$. In the left-hand chart (shape parameter), the limited sample size results in a wide likelihood function. As a result, the

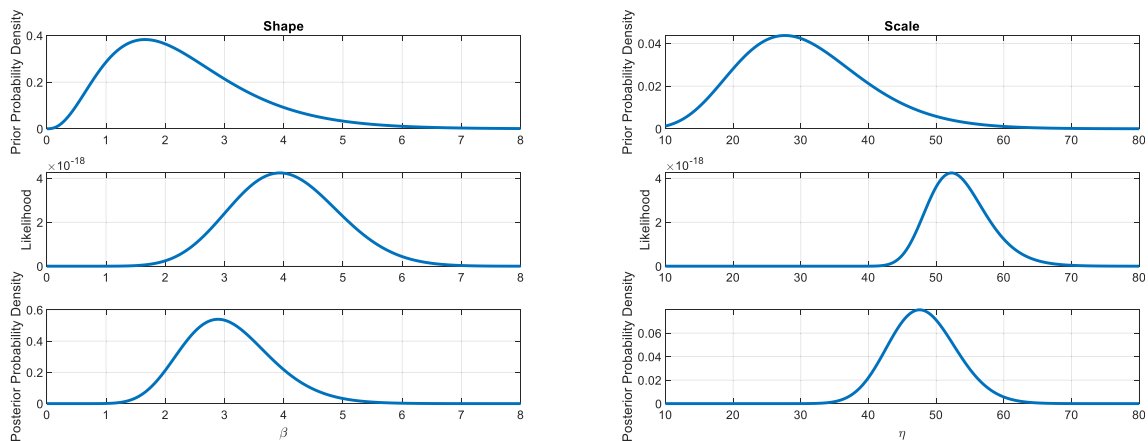


Fig. 7. PLP Plot for Weibull Parameters – 10 samples and priors with high variance.

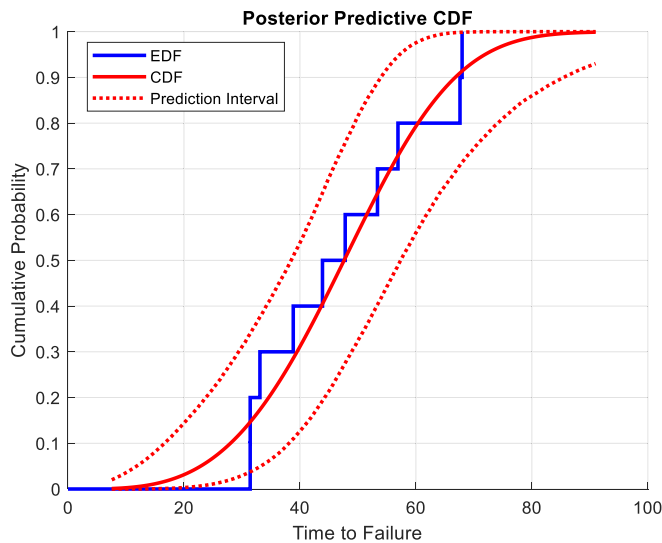


Fig. 8. Example posterior predictive check with 95% PI.

posterior is strongly influenced by the prior. In the right-hand chart (scale parameter), the prior and likelihood have similar variance, resulting in a posterior that lies halfway between the two. The PLP plot in Fig. 6 shows the influence of increasing sample size, whereby the posterior shifts much closer to the maximum likelihood value. In Fig. 7, one can see that priors with higher variance as compared to Figs. 5 and 6 result in increased influence of the likelihoods, even if the likelihoods are wide. (left-hand chart).

In addition to PLP plots, the analyst may wish to examine parameter uncertainty using credible intervals, whereby an $X\%$ credible interval represents the probability that the parameter value lies within the parameter range. We summarize the posterior distribution uncertainty using the central posterior density (CPD) intervals defined as $X\% = (1 - \alpha) \times 100\%$. It is a range of parameter values for which the posterior distribution probability above and below the endpoints is given by $(\alpha/2) \times 100\%$ (Hamada et al., 2008). For example, for a 95% CPD interval, $\alpha = 0.05$.

**These figures are for illustration purposes only. The samples were randomly generated using simulation of a Weibull distribution with $\eta = 50$ and $\beta = 3$. Identical Gamma priors are used for Figs. 5 and 6, and a similar Gamma prior with higher variance was used for Fig. 7.

The posterior predictive distribution is checked by comparing the median CDF and prediction intervals with the empirical distribution function (EDF) as shown in Fig. 8. The EDF is the product limit estimator proposed by Lynden-Bell (1971). The median CDF is calculated using the median values of the Weibull parameters (η and β). Two commonly used

test statistics for goodness-of-fit based on the EDF include the Kolmogorov-Smirnov (KS) test, and the Anderson-Darling (AD) test. The AD test is particularly useful for testing EDF departure from the CDF in the tails of the distribution. Both tests are extensively used in reliability analysis (Tobias and Trindade, 2012).

A visual check of the fitted EDF can be done by comparing where it lies relative to the prediction interval (PI) as shown in Fig. 8. Prediction intervals are calculated by sampling from the posterior distributions of the Weibull parameters and generating a CDF curve for each pair of samples. The $X\% = (1 - \alpha) \times 100\%$ prediction intervals are calculated by considering the set of CDF curves generated and determining the CDF values at the α and $1 - \alpha$ percentiles for each time to failure. For example, for a 95% PI, $\alpha = 0.05$.

4. Case study

Consider, hypothetically, that a decision-maker wishes to prioritize asset replacements using a quantitative risk-based approach. A key component of the risk quantification is the probability of failure of the assets, for which a time to failure distribution is needed. The utility of the proposed methodology for obtaining a time to failure distribution when faced with limited failure data is demonstrated using a case study based on actual experience with assets used in natural gas compression and transmission infrastructure. For confidentiality purposes, the actual data is not presented, rather a simulated data set is used as described in Section 4.1. We examine three cases in this study: inference with no prior information, inference with weakly informative prior information, and inference with prior information elicited from experts. The cases are described in detail in Section 4.2 and inference results are presented in Section 4.3.

4.1. Simulated data

A data set that mimics the actual data is created by simulating the installation and failure of 100 similar assets. The assets were installed in the period between 1955 and 1990. As such, we simulate installation years using a discrete uniform distribution sampling between the installation year range. Random times to failure are generated from a Weibull distribution with parameters $\eta = 30$ and $\beta = 3$. We simulate limited recording of failure data by defining the observation period from 1987 to 2015. Assets installed prior to 1987 are considered truncated samples and assets failing after 2015 are considered suspensions (censored samples). Truncation results in 80 out of the 100 installed assets being considered in the sample for inference (20 having failed at a time before data was recorded). Histograms of the simulated failures and suspensions are plotted with the simulated population pdf in Fig. 9. The complete simulated failure data set is provided in the supplementary materials. A subset of 10 samples is also provided in the supplementary

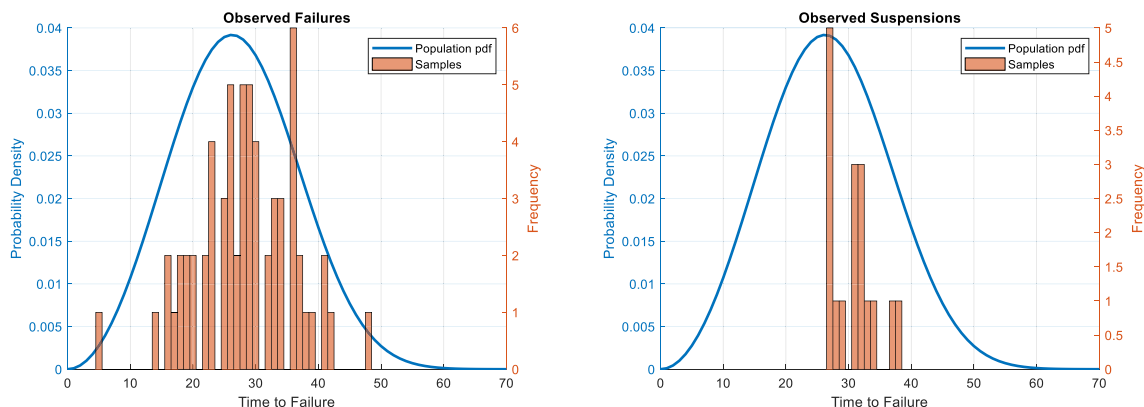


Fig. 9. Simulated Data for Case Study with Weibull population pdf parameters $\eta = 30$ and $\beta = 3$.

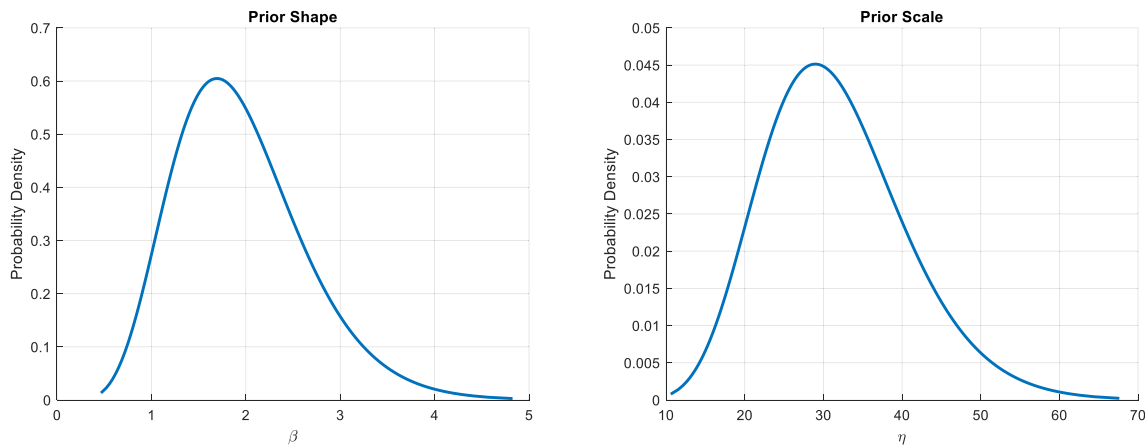


Fig. 10. Prior distributions based on weak prior information.

materials. The subset is used to demonstrate inference with limited samples.

4.2. Prior information cases

In this section we demonstrate the utility of prior information using three case examples. Case 1 demonstrates Bayesian inference using non-informative priors, which is essentially the same as classical inference (e. g. MLE). Case 2 uses weakly informative priors derived from highly uncertain estimates of times to failure. We demonstrate that even weakly informative priors improve the parameter estimates as compared to Case 1. Finally, we demonstrate the power of expert elicited priors in Case 3.

4.2.1. Case 1: non-informative priors

In the first case, Bayesian inference with non-informative priors means that the results will be entirely based on the observed data. This is the Bayesian analogue to classical inference. We define non-informative priors as Gamma distributions with mean of 1 and standard deviation of 100, as proposed by Kundu and Mitra (2016). The choice of hyper-parameters creates a very flat distribution that is essentially non-informative.

4.2.2. Case 2: weakly informative priors

In the second case, to demonstrate the power of priors, we construct informative prior distributions using weak prior information. Weak prior information is given by a $\bar{T}_{mode}$ that is uniformly distributed between 0 and 40, and $\bar{T}_{max}$ that is uniformly distributed between 50 and 100. The choice of uniform distributions and wide ranges (40 and 50 years) is meant to mimic minimal knowledge (high uncertainty) about the asset time to failure. Prior distributions for Weibull shape and scale are generated using the method described in Section 3.3 and shown in Fig. 10. As will be discussed in Section 4.3, even weakly informative priors can greatly improve inference results as compared to inference with non-informative priors.

4.2.3. Case 3: expert elicited priors

Since weakly informative priors can greatly improve the inference results, we expect that priors obtained from experts will improve inference results further, and is modeled appropriately in the third case. Experts are selected and their opinion about asset time to failure is elicited using the methodology described in Section 3.2. Informative priors are constructed based on time to failure estimates elicited from experts as described in Section 3.3.

Before eliciting expert estimates, we obtain an understanding of the failure mechanism and use that understanding to determine the importance of each expert selection criteria; years of experience,

Table 2
AHP criteria comparison matrix.

Comparison Matrix	Years of Experience	Observed Failures	Relevant Training
Years of Experience	1	1/3	5
Observed Failures	3	1	9
Relevant Training	1/5	1/9	1

Table 3
Expert ratings and weights.

Expert	Years of Experience	Observed Failures	Relevant Training	Weight
Expert 1	9	7	1	50.05%
Expert 2	9	5	1	24.63%
Expert 3	5	5	5	17.42%
Expert 4	1	1	9	7.90%

number of observed failures, and level of training with respect to maintenance/operation of the asset in question (see Section 3.2). We know that much of the data is missing because of truncation (i.e. failures occurred before data was recorded). We also know that the degradation cause isn't well understood and that, while in service, the condition of the assets cannot be monitored directly through inspections. The asset is only examined when a removal occurs. As such, we put highest importance on past observations that experts may have had. Given that similar assets are used throughout the natural gas infrastructure, we put strong importance on years of (relevant) experience as we expect that experts with more experience will be able to draw upon knowledge of similar assets that they have observed in operation. It is expected that formal training will not improve an expert's estimate of time to failure because the scientific foundation with respect to the degradation mechanism is limited. As such, it is given the least importance.

Performing pairwise comparison using the rationale described above and the scale presented in Table 1 we evaluate 'Observed Failures' as having an importance of 3 relative to 'Years of Experience', and an importance of 9 relative to 'Relevant Training'. Being consistent with the previous pairwise comparison, 'Years of Experience' is given a relative importance of 5 against 'Relevant Training'. The suggested importance rankings are used to fill in the AHP criteria comparison matrix shown in Table 2.

Once criteria importance is assigned, we rank each expert (on a relative scale) against each AHP criteria. The rankings shown in column 2–3 of Table 3, combined with the importance ratings in Table 2, are used to calculate a weighting factor (Table 3, column 5). More experts

Table 4
Expert estimates for time to failure.

Expert Name	Time to Failure Estimates					
	Mode			Maximum		
	Lower Bound	Best Estimate	Upper Bound	Lower Bound	Best Estimate	Upper Bound
Expert 1	10	20	30	40	50	60
Expert 2	20	30	40	50	60	70
Expert 3	15	20	25	35	40	45
Expert 4	0	20	40	50	75	100

Table 5
Aggregated time to failure estimates.

Estimate	Lower Bound	Best Estimate	Upper Bound
$\bar{T}_{mode}$	12.5	22.5	32.4
$\bar{T}_{max}$	42.4	52.7	63.0

were considered originally but did not score highly in the categories of “Observed Failures” and Years of Experience”, and so had a low relative rank. As such they were excluded from consideration and final weights as shown in Table 3 were calculated for the top four experts. The final weights are applied to the time to failure estimates provided by each expert. Time to failure estimates shown in Table 4 are aggregated using a simple weighted average based on the weights in Table 3. The weighted average results for $\bar{T}_{mode}$ and $\bar{T}_{max}$ are shown in Table 5. Expert rankings

and their estimates have been masked but are based on experience gained through actual interviews with experts with respect to natural gas infrastructure.

Modified Beta-PERT distributions constructed based on the data in Table 5 are shown in Fig. 11. The distribution for $\bar{T}_{mode}$ is given a $\gamma = 4$. The distribution for $\bar{T}_{max}$ is given a $\gamma = 0$. The choice, a subjective interpretation by the analyst, was based on the experts’ feedback that maximum life is difficult to estimate and so their estimates are highly uncertain. Prior distributions generated from the expert input are shown in Fig. 12.

4.3. Results and discussion

Bayesian inference is performed for each case using MH MCMC as described in Section 3.4. We perform inference using a 10-sample data set provided in the supplementary materials. The resultant Weibull parameter estimates are summarized in Table 6 for each case of prior information.

As a point of reference, consider that the resultant parameters using MLE and the limited (10-sample) data set are $\beta = 12.79$ and $\eta = 35.11$. We see that there is little difference between the MLE results and MH MCMC with non-informative priors (Table 6). This is consistent with the findings of McNeish (2016), and underscores the importance of informative priors. With weakly informative priors, the MH MCMC estimate of the Weibull shape parameter is improved as evidenced by the more accurate median estimate and increased precision. However, the estimate of the Weibull scale parameter becomes more uncertain (increased

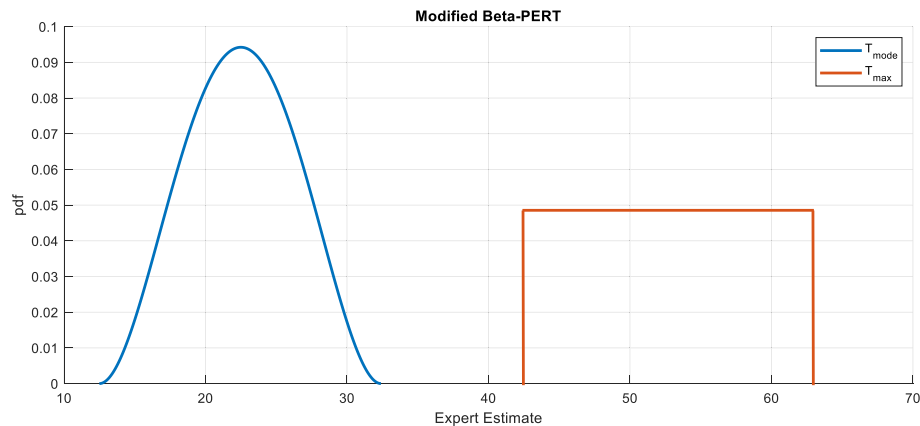


Fig. 11. Estimated Modified Beta-PERT distributions for $\bar{T}_{mode}$ and $\bar{T}_{max}$. Time to failure estimates are summarized in Table 5. The distribution for $\bar{T}_{mode}$ is given a $\gamma = 4$. The distribution for $\bar{T}_{max}$ is given a $\gamma = 0$.

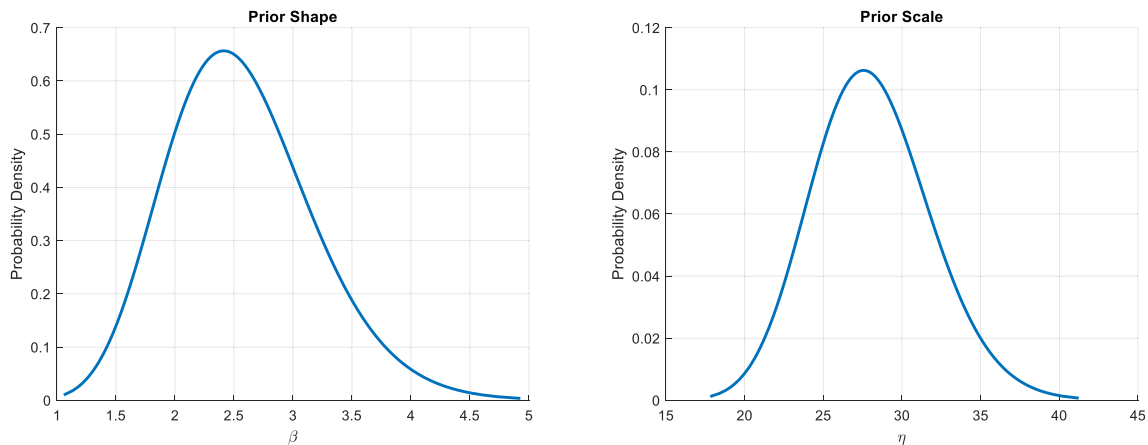


Fig. 12. Prior distributions based on expert estimates.

Table 6Bayesian inference results – 10 samples – true parameters $\eta = 30$ and $\beta = 3$.

Case	Non-Informative Priors	Weakly Informative Priors	Expert Elicited Priors
Shape			
Prior Median	0	1.864	2.495
Posterior Median	9.676	2.648	2.846
with 95% CPD Interval	[3.236,21.67]	[1.378,4.532]	[1.817,4.474]
Scale			
Prior Median	0	30.81	27.91
Posterior Median	36.41	38.73	32.5
with 95% CPD Interval	[31.11,42.28]	[28.56,51.08]	[27.55,39.06]

95% CPD interval), reflecting the uncertainty in the prior information. With expert elicited priors MH MCMC produces superior results for both Weibull parameters. Not only are the estimated parameters more accurate, the precision is increased significantly as compared to the other two cases as shown by the 95% CPD intervals.

Visually comparing (Fig. 13) the predictive distributions resulting from each method of inference, one can see the potential for ill-informed decision-making when faced with limited samples and no prior information. However, even with weak prior information, the difference between the true population life distribution and the inferred distribution can be greatly reduced. Calculating the Kullback-Leibler (KL) divergence between the inferred distribution and the population distribution serves as a means of examining the information lost by the inferred predictive distributions as compared to the population life distribution (Bauckhage, 2013). A KL divergence of 0 is the ideal case (i.e. no information loss). KL values for all the cases are summarized in Table 7, showing significant improvement using expert elicited priors as compared to other means.

Table 8 shows the results of Bayesian inference for each prior information case and with 80 samples. As expected, the results improve (more accurate and precise) with increasing sample size. Moreover, the difference between MH MCMC results and MLE results ($\beta = 3.773$ and $\eta = 31.84$) is almost imperceptible.

The influence of data versus prior information can be visually examined using the PLP plots in Fig. 14 and Fig. 15. As sample sizes increase, the likelihood becomes more precise, causing the posterior distribution to be more influenced by the data. Examining chart for the shape parameter in Fig. 14 we see that with a small sample size the shape parameter value is predominantly based on prior information, underscoring the importance of good prior information.

We have demonstrated (see Table 6) that with limited data both MLE and MH MCMC (using non-informative priors) produce inadequate results. On the other hand, with a large sample size, the difference in the results are virtually identical (see Table 8). The power of prior information is demonstrated using both weakly informative priors and expert elicited priors. Even weakly informative priors improve (i.e. increase accuracy and precision of posteriors) inference results over MLE and MH MCMC with non-informative priors. Further improvement is achieved by using expert elicited priors. By eliciting expert opinion about asset time to failure we can consider knowledge and experience of asset failures (not captured in failure data) through the prior distributions, and augment that knowledge with failure data.

5. Conclusions

This paper presents a methodology to address limited and incomplete data when attempting to perform statistical inference on the time to failure of physical assets. Under the Bayesian paradigm, expert knowledge about asset life is encoded in prior distributions for Weibull parameters. As is shown in this work, incorporation of such knowledge

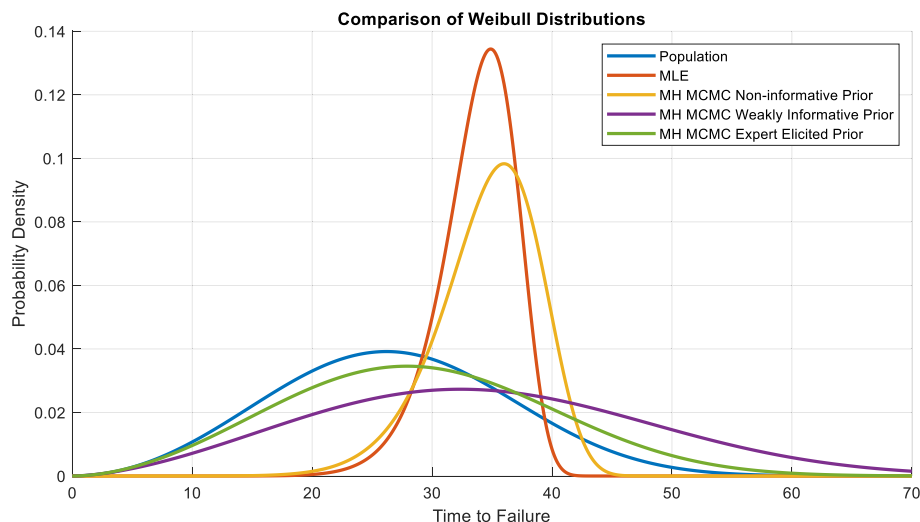
Table 7

KL divergence for inference cases.

Case	KL Divergence
MLE	6.25
MH MCMC with Non-Informative Priors	2.22
MH MCMC with Weakly Informative Priors	0.22
MH MCMC with Expert Elicited Priors	0.031

Table 8Bayesian inference results – 80 samples – true parameters $\eta = 30$ and $\beta = 3$.

Case	Non-Informative Priors	Weakly Informative Priors	Expert Elicited Priors
Shape			
Prior Median	0	1.864	2.495
Posterior Median	3.677	3.390	3.385
with 95% CPD Interval	[2.906,4.574]	[2.694,4.197]	[2.716,4.156]
Scale			
Prior Median	0	30.81	27.91
Posterior Median	31.78	31.55	31.29
with 95% CPD Interval	[29.53,34.15]	[29.16,34.08]	[29.04,33.65]

**Fig. 13.** Posterior predictive comparison.

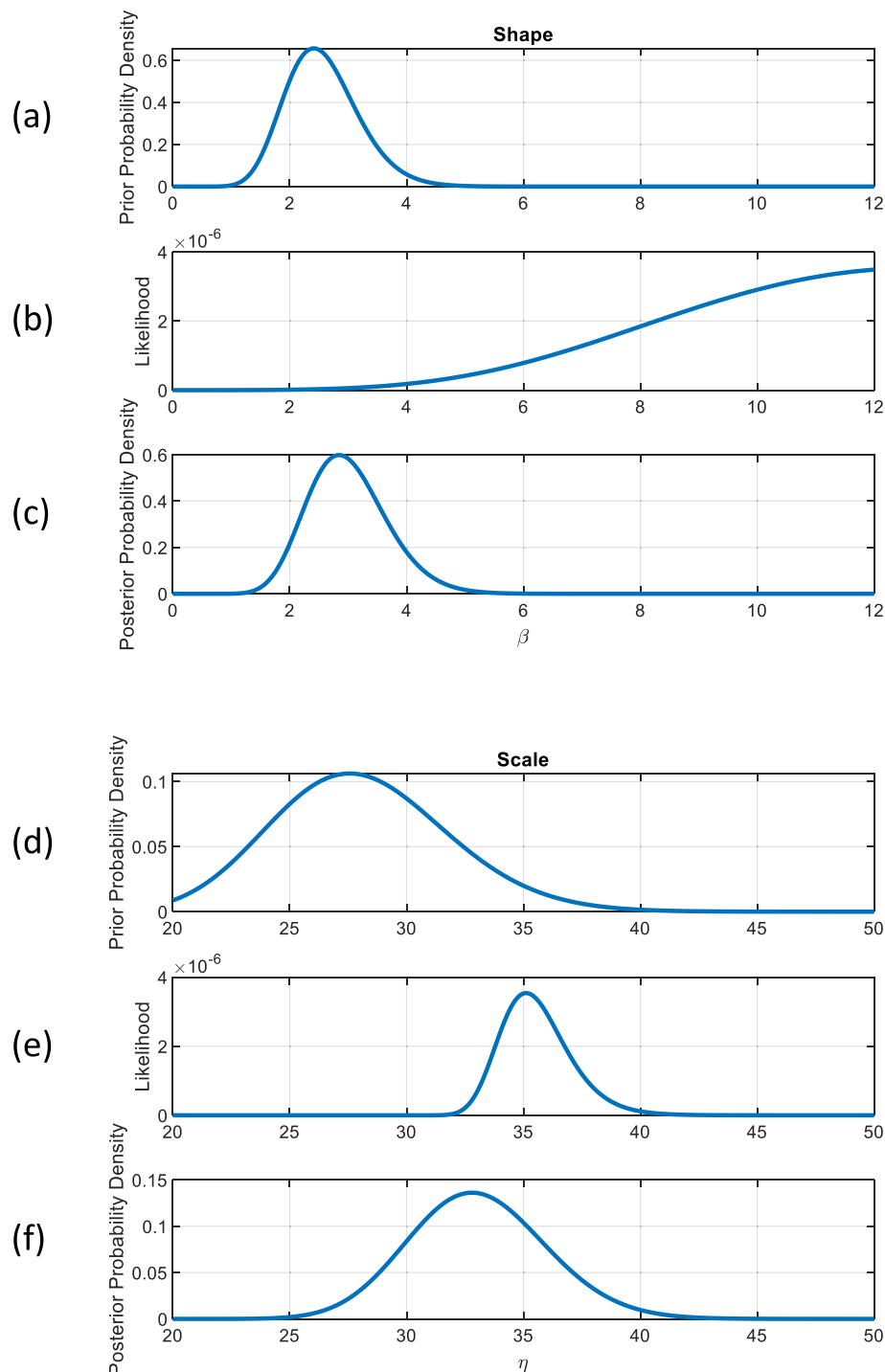


Fig. 14. Prior (a, d), Likelihood (b, e) and Posterior (c, f) Plots for Inference with 10 Samples and Expert Elicited Priors for Weibull shape (a, b, c) and scale (d, e, f) parameters.

may be of crucial importance for proper decision making under uncertainty. AHP is used as a framework for rigorously selecting experts and aggregating their knowledge to construct informative priors. MH MCMC is used to perform Bayesian updating of the prior distributions using actual failure data. We utilize a flexible likelihood function to account for missing or incomplete data (truncation and censoring). Such data issues can occur due to poor data collection practices. As is demonstrated, the result is more accurate estimation of model parameters as well as better precision (reduced uncertainty) of the parameters. The usefulness of the methodology for improved decision making is illustrated using a case study.

Ultimately the results of reliability analysis of assets must be used for decisions regarding design, maintenance and replacement. There are significant costs involved in all such decisions and the decision-maker must be confident in the course of action. As has been illustrated, when dealing with limited, incomplete and missing data, classical methods such as MLE produce inadequate results with a high level of uncertainty. This uncertainty, coupled with the significant capital or operating expenditures at stake, essentially prohibits the decision-maker from properly deciding on a course of action. The methodology presented provides an elegant and powerful way of addressing uncertainty by allowing the analyst to take into consideration all types of available

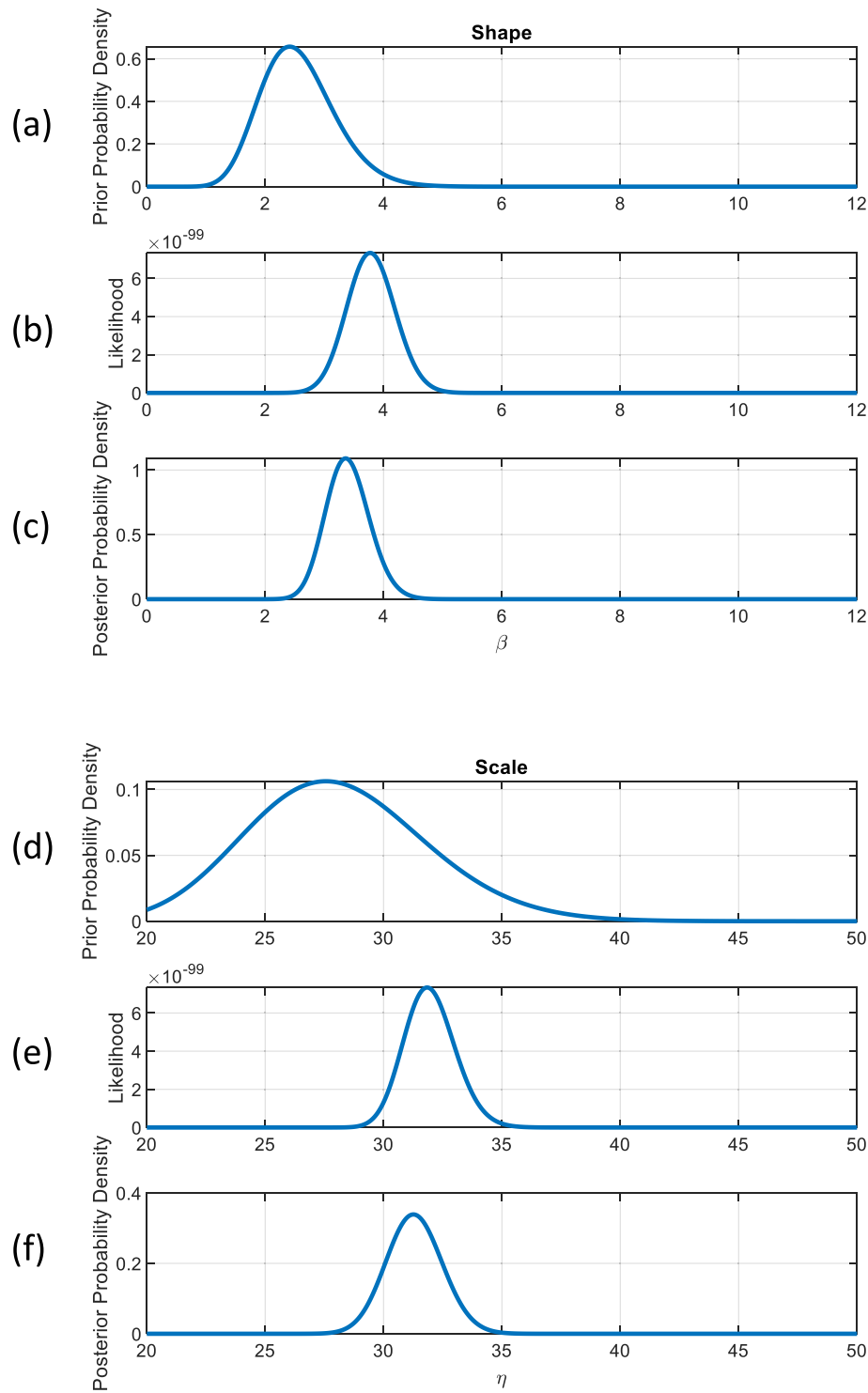


Fig. 15. Prior (a, d), Likelihood (b, e) and Posterior (c, f) Plots for Inference with 80 Samples and Expert Elicited Priors for Weibull shape (a, b, c) and scale (d, e, f) parameters.

information.

While in this paper we have focused on incorporating expert opinion for the construction of priors, the methodology is flexible and can be easily extended to other sources of information. Such information can include accelerated life tests, analogous data from similar assets, data from published reliability databases (e.g. OREDA, HSE HCRD), physics of failure computer simulations, etc. AHP may be used as a means of weighting and aggregating the information from such sources into prior distributions. Future work will explore the suggested approach for

extending the methodology.

Acknowledgements

We thank our reviewers for their careful reading of our manuscript and their many interesting insights and comments.

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jlp.2019.104024>.

References

- Bauckhage, C., 2013. Computing the Kullback-Leibler Divergence between Two Weibull Distributions. Cornell University arXiv:1310.3713 [cs.IT].
- Bedford, T., Cooke, R., 2001. Probabilistic Risk Analysis: Foundations and Methods. Cambridge University Press, Cambridge.
- Bedford, Tim, Quigley, John, Walls, Lesley, 2006. Expert elicitation for reliable system design. *Stat. Sci.* 21 (4), 428–450.
- Bentley, John, 1999. Introduction to Reliability and Quality Engineering, second ed. Pearson Education Limited, Harlow, England.
- Cagno, E., Caron, F., Mancini, M., Ruggeri, F., 2000. Using AHP in determining the prior distributions on gas pipeline failures in a robust Bayesian approach. *Reliab. Eng. Syst. Saf.* 67, 275–284.
- Chib, Siddhartha, Greenberg, Edward, 1995. Understanding the Metropolis-Hastings Algorithm. *American Statistician*, pp. 327–335.
- Coolen, F.P.A., Mertens, P.R., Newby, M.J., 1990. A Bayes-Competing Risk Model for the Use of Expert Judgement in Reliability Estimation. Memorandum COSOR, vol. 9032. Technische Universiteit Eindhoven, Eindhoven.
- Duben, A.J., Ricks, J., 2015. The application of the analytic hierarchy process in sales personnel recruitment. In: Proceedings of the 1992 Academy of Marketing Science Annual Conference. Springer, Cham, pp. 417–421.
- Emura, Takeshi, Shiu, Shau-Kai, 2014. Estimation and Model Selection for Left-Truncated and Right Censored Lifetime Data with Application to Electric Power Transformers Analysis. Graduate Institute of Statistics, National Central University, Taiwan.
- Evans, J.S.T.B., 1989. Bias in Human Reasoning: Causes and Consequences. Erlbaum Associates, Inc, Hillsdale, NJ.
- Fengqiang, Gong, Li, Xibing, Deng, Jian, 2006. Probabilistic distribution of geotechnical parameters by using AHP prior distribution fusion method. *Chin. J. Geotech. Eng.* 28 (10), 1313–1318.
- Gelman, Andrew, Rubin, Donald, 1992. Inference from iterative simulation using multiple sequences. *Stat. Sci.* 457–472.
- Gutierrez-Pulido, H., Aguirre-Torres, V., Christen, J Andres, 2005. A practical method for obtaining prior distributions in reliability. *IEEE Trans. Reliab.* 54 (2), 262–269.
- Hamada, Michael S., Wilson, Alyson G., Reese, C Shane, Martz, Harry F., 2008. Bayesian Reliability. Springer, New York, NY.
- Hong, By Yili, Meeker, William, McCalley, James, 2009a. Prediction of remaining life of power transformers based on left truncated and right censored lifetime data. *Ann. Appl. Stat.* 857–879.
- Hong, Y., Meeker, W.Q., McCalley, J.D., 2009b. Prediction of remaining life of power transformers based on left truncated and right censored lifetime data. *IEEE Trans. Reliab.* 3, 857–879.
- Jardine, Andrew KS., Tsang, Albert HC., 2013. Maintenance, Replacement, and Reliability Theory and Applications, second ed. CRC Press, Boca Raton, FL.
- Johnson, S.R., Tomlinson, G.A., Hawker, G.A., Granton, J.T., Feldman, B.M., 2010. Methods to elicit beliefs for Bayesian priors: a systematic review. *J. Clin. Epidemiol.* 63, 355–369.
- Knox, Steven W., 2018. Machine Learning: a Concise Introduction. John Wiley & Sons, Inc, Hoboken, NJ.
- Kundu, Debasis, Mitra, Debanjan, 2016. Bayesian inference of Weibull distribution based on left truncated and right censored data. *Comput. Stat. Data Anal.* 99, 38–50.
- Lad, Bhupesh K., Kulkarni, Makarand S., 2010. A parameter estimation method for machine tool reliability analysis using expert judgement. *Int. J. Data Anal. Tech. Strateg.* 2 (2), 155–169.
- Lambert, B., 2018. A Student's Guide to Bayesian Statistics. SAGE Publications, Inc, Thousand Oaks, CA.
- Li, Ming, Meeker, William Q., 2014. "Application of Bayesian Methods in Reliability Data Analyses." *Journal of Quality Technology*. Iowa State University, pp. 1–23.
- Lynden-Bell, D., 1971. A method of allowing for known observational selection in small samples applied to 3CR quasars. *Mon. Not. R. Astron. Soc.* 155, 95–118.
- McNeish, Daniel, 2016. On using Bayesian methods to address small sample problems. *Struct. Equ. Model.: A Multidiscip. J.* 23 (5), 750–773.
- Mitra, Debanjan, 2012. Likelihood Inference for Left Truncated and Right Censored Lifetime Data. McMaster University, Hamilton.
- Pasman, H., Reniers, G., 2014. Past, present and future of quantitative risk assessment (QRA) and the incentive it obtained from land-use planning (LUP). *J. Loss Prev. Process. Ind.* 28, 2–9.
- Quigley, J., Bedford, T., Walls, L., 2008. Prior distribution elicitation. In: *Encyclopedia of Statistics in Quality and Reliability*, by F Ruggeri, RS Kenett and FW Faltin.
- Redman, Thomas C., 2018. If your data is bad, your machine learning tools are useless. *Harv. Bus. Rev.* <https://hbr.org/2018/04/if-your-data-is-bad-your-machine-learning-tools-are-useless>.
- Saaty, Thomas L., 1980. The Analytic Hierarchy Process: Planning, Priority Setting, Resource Allocation. McGraw-Hill International Book Co, New York.
- Saaty, Thomas L., Vargas, Luis G., 2012. Models, Methods, Concepts & Applications of the Analytic Hierarchy Process. Springer, New York.
- Singpurwalla, Nozer D., 1988. An interactive PC-based procedure for reliability assessment incorporating expert opinion and survival data. *J. Am. Stat. Assoc.* 83 (401), 43–51.
- Sridhar, N., 2018. Knowledge-based predictive analytics in corrosion. *Corrosion* 74, 181–196.
- Tobias, Paul A., Trindade, David C., 2012. Applied Reliability, third ed. CRC Press, Boca Raton, FL.
- Tversky, A., Kahneman, D., 1982. A Judgement under Uncertainty: Heuristics and Biases. Cambridge University Press, Cambridge.
- van de Schoot, R., Denissen, J., Neyer, F.J., Kaplan, D., Assendorp, J.B., MA van Aken, 2013. A gentle introduction to bayesian analysis: applications to developmental research. *Child Dev.* 1–19.
- Villa, Valeria, Paltrinieri, Nicola, Khan, Faisal, Cozzani, Valerio, 2016. Towards dynamic risk analysis: a review of the risk assessment approach and its limitations in the chemical process industry. *Saf. Sci.* 89, 77–93.
- Vose, D., 2008. Risk Analysis - A Quantitative Guide, third ed. John Wiley & Sons Ltd, Chichester, West Sussex.
- Wilson, Alyson, Fronczyk, Kassandra, 2016. Bayesian Reliability: Combining Information. Institute for Defense Analysis, Alexandria.
- Wu, JenHao, 2017. Reliability Analysis for Small Wind Turbines Using Bayesian Hierarchical Modeling. The University of Edinburgh, Edinburgh.